Finite Part of Wilson–Dirac Relative Entropy

Kirtus Justus (Theoretical Physicist)
Port Coquitlam, British Columbia, Canada

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Abstract

We determine the finite part of the relative entropy between massive and massless infinite-chain Wilson–Dirac occupied-band states restricted to an interval. For interval length L, bare mass $m = \mu/L$ and fixed $\mu > 0$, the result is $\mathcal{D}_L(\mu) = \mu^2 \log L/6 + \mu^2(13/36 - \pi/12 - \log \mu/6) - \mathcal{S}(\mu) + O_\mu(1/\log L)$, where $\mathcal{S}$ is a finite boundary entropy trace path. Two estimates supply the additional convergence needed beyond the leading logarithm. A Riccati comparison gives the global half-filled modular bound $\|E_N\| \le 4\sqrt{N}$ and a positive-majorant weighted error $O_\mu(N^{-1/2})$. An elementary estimate for Jacobi paths at every degree proves convergence of the nonlinear trace, with error $O_\mu((\log N/N)^{1/3})$. The proof uses the previously established exact concentration connection, fixed-interior expansion and boundary entropy limit, and states these dependencies explicitly. No uniform moving-edge expansion is asserted.

1 Wilson comparison and main result

Let σ_x, σ_z be Pauli matrices and define

$$h_m(k) = \sin k\, \sigma_x + (1 + m - \cos k)\sigma_z, \qquad P_m(k) = \frac{1}{2}\left[I - \frac{h_m(k)}{\sqrt{m^2 + 4(1+m)\sin^2(k/2)}}\right].$$

The value of P_0 at the isolated degeneracy is irrelevant to Fourier multiplication. Its compression to L consecutive two-orbital cells is

$$(C_{L,m})_{ab} = \frac{1}{2\pi}\int_{-\pi}^{\pi} e^{ik(a-b)} P_m(k)\, \mathrm{d}k, \qquad 0 \le a, b < L.$$

For finite positive contraction matrices, write $S(C) = -\operatorname{Tr}[C \log C + (I - C)\log(I - C)]$, with $0 \log 0 = 0$. For $0 < A, B < I$, set

$$\mathcal{D}(A\|B) = \operatorname{Tr}\{A(\log A - \log B) + (I - A)[\log(I - A) - \log(I - B)]\}.$$

The massive covariance is the first argument. Put $\mathcal{D}_L(\mu) = \mathcal{D}(C_{L,\mu/L}\|C_{L,0})$. These are restrictions of infinite-chain projectors; they are not open-chain ground-state projections.

Theorem 1 (Finite relative-entropy part)**.** *For every fixed $\mu > 0$, as integer $L \to \infty$,*

$$\boxed{\mathcal{D}_L(\mu) = \frac{\mu^2}{6}\log L + \mu^2\left(\frac{13}{36} - \frac{\pi}{12} - \frac{\log \mu}{6}\right) - \mathcal{S}(\mu) + O_\mu((\log L)^{-1}).} \tag{1}$$

The finite boundary quantity $\mathcal{S}(\mu)$ is defined by the trace path in Section 7. The error constant and asymptotic onset may depend on μ. The displayed rate is not claimed to be sharp.

The finite covariance identity and geometry are those of [1, 2, 3]. Sections 2–4 prove a global modular estimate from the exact differential connection in [5, 6, 7]. Sections 5–6 prove the nonlinear limit from finite paths and the positive Wilson majorant. The final assembly imports the two-bond expansion [2] and the boundary entropy theorem [10, 11]. No finite-part convergence hypothesis is used.

2 Model and exact inputs

Let $N \geq 2$ be even and set $L = N/2$. For $0 \leq j, k < N$, define

$$B_{jk} = \begin{cases} 1/2, & j = k, \\ \sin[\pi(j-k)/2]/[\pi(j-k)], & j \neq k, \end{cases} \qquad J_{j-1,j} = J_{j,j-1} = \frac{j(N-j)}{N} \quad (1 \leq j < N),$$

with $J_{jj} = 0$. We also write $B_N = B$ and $J_N = J$. Let U_N be the interval-preserving unitary of [2] and set $A_N = U_N C_{L,\mu/L} U_N^*$, so that $B_N = U_N C_{L,0} U_N^*$. The matrices B_N and J_N commute, J has simple spectrum, $0 < B < I$, and $|\lambda/N| < 1/2$ for every Jacobi eigenvalue. Write

$$K_N = \log(I - B) - \log B, \qquad E_N = K_N + N\eta(J/N),$$

where

$$\eta(t) = \pi t \, {}_3F_2(1/4, 1/2, 3/4; 1, 3/2; 4t^2).$$

At a common mode, $Jv = \lambda v$, $Bv = \zeta v$, put $t = \lambda/N$ and $e = \log[(1-\zeta)/\zeta] + N\eta(t)$.

We use the exact whole-line construction of [5], whose operator-domain and finite-spectrum connection inputs are established in [6, 7]. Set

$$h = N^{-1}, \qquad p(r) = \operatorname{sech} r, \qquad Q_t(r) = \frac{1 - 2tp(r)}{4}, \qquad v(r) = \frac{p(r)^2}{4}.$$

For $|t| < 1/2$, the left recessive solution of $[-h^2 D_r^2 + Q_t + h^2 v]u = 0$ gives the positive logarithmic derivative

$$w_t = hu'/u, \qquad hw_t' + w_t^2 = Q_t + h^2 v, \qquad w_t(-\infty) = 1/2. \tag{2}$$

The exact spectral interpolation $\mathcal{F}_N$ satisfies

$$f_N(t) := \frac{\mathcal{F}_N(t)}{N} = \int_{\mathbb{R}} (w_{-t} - w_t) \, \mathrm{d}r, \qquad \mathcal{F}_N(\lambda/N) = \log \frac{\zeta}{1-\zeta}, \tag{3}$$

$$\eta(t) = \int_{\mathbb{R}} (s_{-t} - s_t) \, \mathrm{d}r, \qquad s_t := \sqrt{Q_t}. \tag{4}$$

Both f_N and η are odd. Individual differences $w_t - 1/2$ and $s_t - 1/2$ are integrable for each $|t| < 1/2$.

If $G_h(t; r, s)$ denotes the Green kernel of $P_h(t) = -h^2 D_r^2 + Q_t + h^2 v$, the weighted resolvent identity gives

$$f_N'(t) = \frac{h}{2} \int_{\mathbb{R}} p(r) \big[G_h(t; r, r) + G_h(-t; r, r) \big] \, \mathrm{d}r. \tag{5}$$

For any positive lower bound m on s_t, constant Riccati barriers give $w_t \geq m$, and reduction of order gives

$$0 < G_h(t; r, r) = \frac{1}{h^2} \int_r^\infty \exp\left[-\frac{2}{h} \int_r^a w_t(b) \, \mathrm{d}b \right] \mathrm{d}a \leq \frac{1}{2hm}. \tag{6}$$

These identities hold at every interior parameter; no limiting formula at $t = 1/2$ will be required.

3 An elementary global estimate

Lemma 2 (Leading square-root comparison)**.** *For $|t| < 1/2$, let $m_t = \inf_{r \in \mathbb{R}} s_t(r) > 0$. Then*

$$\| w_t - s_t \|_{L^1(\mathbb{R})} \leq \frac{h \operatorname{Var}(s_t) + h^2/2}{2m_t}. \tag{7}$$

Here, for $0 \leq t < 1/2$,

$$m_t = \tfrac{1}{2}\sqrt{1 - 2t}, \qquad \operatorname{Var}(s_t) = 1 - \sqrt{1 - 2t},$$
$$m_{-t} = \tfrac{1}{2}, \qquad \operatorname{Var}(s_{-t}) = \sqrt{1 + 2t} - 1. \tag{8}$$

Proof. The exact difference $d_t = w_t - s_t$ satisfies

$$hd_t' + (w_t + s_t)d_t = h^2 v - hs_t', \qquad d_t(-\infty) = 0.$$

Since $w_t + s_t \geq 2m_t$, its integrating-factor representation yields

$$|d_t(r)| \leq \frac{1}{h} \int_{-\infty}^{r} |h^2 v(a) - hs_t'(a)| e^{-2m_t(r-a)/h} \, \mathrm{d}a.$$

Tonelli's theorem and $\int_{\mathbb{R}} v \, \mathrm{d}r = 1/2$ prove (7). The potential has one extremum, at $r = 0$, and tends to $1/4$ at both ends. This gives (8). In particular, for the negative parameter, the infimum remains $1/4$ for Q_{-t} at infinity; its central value $(1 + 2t)/4$ is a maximum, not a lower bound. □

Lemma 3 (A shared integrable derivative bound)**.** *For $0 \leq t < 1/2$,*

$$0 \leq f_N'(t), \eta'(t) \leq M(t) := \frac{\pi}{2} \left[(1 - 2t)^{-1/2} + 1 \right]. \tag{9}$$

The upper bound is independent of N and integrable up to $1/2$.

Proof. Use $m_t = \sqrt{1 - 2t}/2$ and $m_{-t} = 1/2$ separately in (6), then substitute in (5) and use $\int_{\mathbb{R}} p \, \mathrm{d}r = \pi$. For (4), differentiation on any fixed compact interior parameter interval is justified by an integrable multiple of p, and gives

$$\eta'(t) = \frac{1}{4} \int_{\mathbb{R}} p(r) \left(\frac{1}{s_{-t}(r)} + \frac{1}{s_t(r)} \right) \mathrm{d}r.$$

The same lower bounds yield (9). □

Theorem 4 (Global sublinear discrepancy)**.** *For every even $N \geq 2$,*

$$\sup_{|t| < 1/2} |f_N(t) - \eta(t)| \leq 4N^{-1/2}, \qquad \boxed{\|E_N\| \leq 4\sqrt{N}.} \tag{10}$$

Proof. Set $h = 1/N \leq 1/2$ and $\tau_h = (1 - h)/2$. For $0 \leq t \leq \tau_h$, one has $2m_t \geq \sqrt{h}$, $\mathrm{Var}(s_t) \leq 1$ and $\mathrm{Var}(s_{-t}) \leq \sqrt{2} - 1 < 1/2$. Thus (7) gives

$$\|w_t - s_t\|_1 \leq \sqrt{h} + \tfrac{1}{2} h^{3/2}, \qquad \|w_{-t} - s_{-t}\|_1 \leq \tfrac{1}{2} h + \tfrac{1}{2} h^2.$$

By (3)–(4), the interior error is at most

$$R_h = \sqrt{h} + \tfrac{1}{2} h + \tfrac{1}{2} h^{3/2} + \tfrac{1}{2} h^2. \tag{11}$$

For $\tau_h < t < 1/2$, both increments $f_N(t) - f_N(\tau_h)$ and $\eta(t) - \eta(\tau_h)$ lie in $[0, U_h]$, where

$$U_h = \int_{\tau_h}^{1/2} M(s) \, \mathrm{d}s = \frac{\pi}{2} \left(\sqrt{h} + \frac{h}{2} \right).$$

Consequently their difference has absolute value at most U_h, and $|f_N(t) - \eta(t)| \leq R_h + U_h$. This controls the full open edge interval without evaluating either function at its endpoint. Oddness gives the same estimate for negative t.

Finally,

$$\frac{R_h + U_h}{\sqrt{h}} = 1 + \frac{\pi}{2} + \left(\frac{1}{2} + \frac{\pi}{4} \right) \sqrt{h} + \frac{h}{2} + \frac{h^{3/2}}{2} \leq \frac{445}{112} < 4.$$

For the rational bound use $\pi < 22/7$, $\sqrt{h} \leq 3/4$, $h \leq 1/2$ and $h^{3/2} \leq 3/8$. At each finite mode, (3) gives $e = N[\eta(t) - f_N(t)]$. The common orthonormal eigenbasis of B and J therefore converts the scalar bound into (10). □

4 Positive-majorant and Wilson consequences

Corollary 5 (Vanishing positive weighted error)**.** *Suppose $G_N \geq 0$ and fixed constants $M, a, b \geq 0$ satisfy*

$$\operatorname{Tr} G_N \leq M, \qquad \nu_N((x,\infty)) := \operatorname{Tr}[G_N \mathbf{1}_{\{|J|>x\}}] \leq a/x + b/x^2 \quad (x>0). \tag{12}$$

There exist universal finite C, N_0 such that

$$\boxed{\operatorname{Tr}(G_N|E_N|) \leq \frac{CM}{N} + \frac{16a}{\sqrt{N}} + \frac{64b}{N^{3/2}} \quad (N \geq N_0,\ N \text{ even}).} \tag{13}$$

If $-G_N \leq S_N \leq G_N$, the same upper bound controls $|\operatorname{Tr}(S_N E_N)|$. No commutation of G_N or S_N with J is assumed.

Proof. The fixed-interior theorem of [5], with $K=1$ and $\tau = 1/4$, gives $|e| \leq C/N$ for every common mode with $|\lambda| \leq N/4$, once $N \geq N_0$. Sum in this common eigenbasis. On the complement use (10) and (12):

$$\operatorname{Tr}(G_N|E_N|) \leq CM/N + 4\sqrt{N}\,\nu_N((N/4,\infty)) \leq CM/N + 16aN^{-1/2} + 64bN^{-3/2}.$$

For each basis vector v_j, the order bounds imply $|\langle v_j, S_N v_j\rangle| \leq \langle v_j, G_N v_j\rangle$. Hence

$$|\operatorname{Tr}(S_N E_N)| \leq \sum_j |e_j|\,|\langle v_j, S_N v_j\rangle| \leq \operatorname{Tr}(G_N|E_N|).$$

This uses scalar diagonal expectations, not the generally invalid operator implication $|S_N| \leq G_N$. □

For the Wilson interval of L two-orbital cells, $N = 2L$, $m = \mu/L$, and fixed $\mu > 0$, the exact decomposition and majorant theorem of [3] give

$$A_N - B_N = S_N + D_N^{\mathrm{tr}}, \qquad M = \frac{\mu}{\pi}, \quad a = \sqrt{\frac{2}{15}}\frac{\mu^2}{\pi}, \quad b = \frac{\mu}{3\pi}.$$

Thus

$$\operatorname{Tr}(G_N|E_N|) = O_\mu(N^{-1/2}), \qquad \boxed{\operatorname{Tr}(S_N E_N) = O_\mu(N^{-1/2}) \longrightarrow 0.} \tag{14}$$

The already established occupation-weighted variance and transverse pairing estimate in [5, 3] give

$$|\operatorname{Tr}(D_N^{\mathrm{tr}} E_N)| = O_\mu(\sqrt{\log N}/N).$$

Consequently the full covariance difference satisfies

$$\operatorname{Tr}[(A_N - B_N)E_N] = O_\mu(N^{-1/2}) \longrightarrow 0. \tag{15}$$

5 Jacobi paths at every degree

Write $T_N = J_N/N$, $r = \mu/(N+\mu)$ and

$$F(t) = \eta(t) - \pi t = \sum_{n\geq 1} a_n t^{2n+1}, \qquad a_n = \frac{\pi \binom{2n}{n}\binom{4n}{2n}}{(2n+1)16^n}.$$

Here F denotes only this scalar nonlinear function, not the logit interpolation $\mathcal{F}_N$. The longitudinal covariance is exactly the Toeplitz matrix of

$$s_r(q) = -\operatorname{sgn}(\cos q)\, g_r(q), \qquad g_r(q) = \frac{1}{2}\left(1 - \frac{|\cos q|}{\sqrt{\cos^2 q + r^2 \sin^2 q}}\right). \tag{16}$$

This follows from the interval-preserving scalarization of [3, 4]. Its coefficients are real and even, vanish at even displacements, and for odd d obey

$$s_d = -\frac{1}{\pi}\int_0^{\pi/2} \left(1 - \frac{\cos q}{\sqrt{\cos^2 q + r^2 \sin^2 q}}\right)\cos(dq)\,\mathrm{d}q.$$

In particular $s_1 < 0$. For a periodic function u, put $\langle u \rangle = (2\pi)^{-1} \int_{-\pi}^{\pi} u(q)\,\mathrm{d}q$. For odd d, $|\cos(dq)| \le |d||\cos q|$, by the identity for $\sin(d\theta)/\sin\theta$ with $q = \pi/2 - \theta$. A split at $|\cos q| = r$ gives

$$J_1(r) := \langle g_r |\cos q| \rangle \le \frac{r^2}{\pi}\left(1 + \frac{1}{2}\log\frac{2}{r}\right), \qquad |s_d| \le |d| J_1(r), \quad 0 < r \le 1/2. \tag{17}$$

Indeed $0 \le 1 - (1+x)^{-1/2} \le \min(1, x/2)$; the two quadrant integrals are bounded using $\int_{\arccos r}^{\pi/2} \cos q\,\mathrm{d}q \le r^2$ and $\int_0^{\arccos r} \sec q\,\mathrm{d}q \le \log(2/r)$.

Lemma 6 (Uniform finite-path estimate)**.** *Set $I_p = (p!)^2/(2p+1)!$ and $P_{p,N}(q) = \sum_{i,j=0}^{N-1} (T_N^p)_{ij} e^{iq(j-i)}$. For every $N \ge 2$, every odd $p \ge 1$, and every real q,*

$$|P_{p,N}(q) - N I_p (2\cos q)^p| \le (4p^3 + 2) 2^{-p} |\cos q|. \tag{18}$$

There is no restriction $p < N$.

Proof. Extend $\beta(x) = x(1-x)$ by zero outside $[0,1]$ and call the extension β_+. It is 1-Lipschitz and bounded by $1/4$. A length-p nearest-neighbor path starting at i has weight $w_i = \prod_{\ell=1}^{p} \beta_+((i+b_\ell)/N)$, where $|b_\ell| \le p$. A step from displacement a uses bond offset $a+1$ when positive and a when negative. Any path leaving the interval crosses bond 0 or N, so its zero-extended weight is exactly zero. Thus unrestricted path summation gives the finite matrix power, including degrees $p \ge N$. Telescoping the product yields

$$|w_i - \beta_+(i/N)^p| \le p^2 4^{1-p}/N.$$

Summing over the N starts and 2^p paths bounds the Fourier coefficient ℓ^1 error by $4p^2 2^{-p}$. Exact and approximate coefficients are symmetric in displacement. Pair the coefficients at d and $-d$ and use odd parity and $|\cos(dq)| \le p|\cos q|$ to obtain $4p^3 2^{-p}|\cos q|$. Finally β_+^p has total variation $2 \cdot 4^{-p}$, so

$$\left| \sum_{i=0}^{N-1} \beta(i/N)^p - N \int_0^1 \beta(x)^p\,\mathrm{d}x \right| \le 2 \cdot 4^{-p}.$$

The Euler beta integral [12] is I_p. Multiplying this common Riemann-sum error by $(2\cos q)^p$ contributes at most $2^{1-p}|\cos q|$. Adding the two estimates proves the claim. □

The exact Toeplitz trace identity is $\mathrm{Tr}(S_N T_N^p) = \langle s_r P_{p,N} \rangle$. Consequently

$$\left| -N \,\mathrm{Tr}(S_N T_N^p) + N^2 I_p \langle s_r (2\cos q)^p \rangle \right| \le N(4p^3+2) 2^{-p} J_1(r). \tag{19}$$

For each fixed odd $p \ge 3$, dominated convergence in this formula gives

$$-N \,\mathrm{Tr}(S_N T_N^p) \longrightarrow \mu^2 \kappa_p, \qquad \kappa_p = \frac{2^{p-1} I_p}{\pi} \int_0^{\pi/2} \sin^2 q \cos^{p-2} q\,\mathrm{d}q. \tag{20}$$

The dominating angular function is a constant times $\sin^2 q |\cos q|^{p-2}$, because $|s_r|/r^2 \le \sin^2 q/(4\cos^2 q)$. The path error is $O_{p,\mu}(\log N/N)$. At $p = 3$, $\kappa_3 = 1/(105\pi)$.

6 The nonlinear finite part and its rate

Let $\mathcal{Q}_N = -N \,\mathrm{Tr}[(S_N + D_N^{\mathrm{tr}}) F(T_N)]$ and $C_\eta = 8F(1/2) < \infty$. Euler's beta integral gives the exact identity

$$a_n \kappa_{2n+1} = \frac{1}{2n(4n+1)(4n+2)(4n+3)}, \qquad c_\eta := \sum_{n \ge 1} a_n \kappa_{2n+1} = \frac{11}{18} - \frac{\pi}{12} - \frac{\log 2}{2}. \tag{21}$$

For completeness, the sum equals $-\frac{1}{4}\int_0^1 (1-x)^2 \log(1-x^4)\,\mathrm{d}x$. Integration by parts makes this $\frac{1}{3}\int_0^1 x^3(1-x)^2/[(1+x)(1+x^2)]\,\mathrm{d}x$; polynomial division gives $x^2 - 3x + 3 - 2/(x+1) + (x-1)/(x^2+1)$ inside the latter integral. This proves the stated constant. Its positive tail is bounded by $\sum_{n>K} a_n \kappa_{2n+1} \le 1/(384K^3)$.

The central-binomial bound $\binom{2n}{n} \le 4^n/\sqrt{\pi n}$ follows by comparing $\int_0^{\pi/2} \cos^{2n} x\,\mathrm{d}x$ with $\int_0^\infty e^{-nx^2}\,\mathrm{d}x$. It implies

$$a_n 2^{-(2n+1)} \le \frac{1}{2\sqrt{2}\,n(2n+1)}, \qquad |F(t) - F_K(t)| \le \frac{\sqrt{2}}{K}|t|^3, \tag{22}$$

where $F_K = \sum_{n=1}^K a_n t^{2n+1}$ and $|t| \le 1/2$. The Wilson majorant tail implies

$$M_{3,\mu,N} := \frac{1}{N^2}\operatorname{Tr}(G_N|J_N|^3) \le \frac{3a_\mu}{8} + \frac{3b_\mu}{2N}, \quad a_\mu = \sqrt{2/15}\frac{\mu^2}{\pi}, \quad b_\mu = \frac{\mu}{3\pi}. \tag{23}$$

This is layer-cake integration up to $N/2$. Diagonal expectations of $-G_N \le S_N \le G_N$ and (22) therefore give

$$|N\operatorname{Tr}\{S_N(F - F_K)(T_N)\}| \le \frac{\sqrt{2}}{K}\left(\frac{3a_\mu}{8} + \frac{3b_\mu}{2N}\right). \tag{24}$$

It justifies the iterated fixed-degree limit without exchanging uncontrolled infinite sums and limits.

For the rate, put $c = |\cos q|$, $\epsilon = r^2 \sin^2 q/c^2$ and

$$J_3(r) = \langle c^3(\epsilon/4 - g_r)\rangle.$$

Taylor's theorem gives $0 \le \epsilon/4 - g_r \le \min(\epsilon/4, 3\epsilon^2/16)$. Splitting at $c = r$, as in (17), yields

$$J_3(r) \le \frac{r^4}{\pi}\left(\frac{1}{2} + \frac{3}{8}\log\frac{2}{r}\right). \tag{25}$$

The split is essential for the borderline cubic term: a bound proportional to $1/c$ cannot be integrated down to $c = 0$. Because $F_K(t) \le C_\eta t^3$ for $0 \le t \le 1/2$, the symbol replacement error in the finite polynomial phase integral is at most

$$N^2 C_\eta \int_0^1 [2x(1-x)]^3\,\mathrm{d}x\, J_3(r) = \frac{2C_\eta}{35}N^2 J_3(r).$$

For $p = 2n+1$, (22) also gives

$$a_n(4p^3 + 2)2^{-p} \le \frac{4p^2 + 2/p}{2\sqrt{2}\,n} \le \frac{19}{\sqrt{2}}n,$$

using $p \le 3n$ and $2/(pn) \le 2n/3$. Thus the sum through K is at most $19K(K+1)/(2\sqrt{2})$. Finally, $|N^2r^2 - \mu^2| \le 2\mu^3/N$.

Theorem 7 (Quantitative nonlinear convergence)**.** *For even $N \ge \max(2,\mu)$ and every integer $K \ge 1$,*

$$\begin{aligned}|\mathcal{Q}_N - \mu^2 c_\eta| \le{}& \frac{19}{2\sqrt{2}}NJ_1(r)K(K+1) + \frac{2C_\eta}{35}N^2J_3(r) + \frac{2c_\eta\mu^3}{N} + \frac{\mu^2}{384K^3}\\ &+ \frac{\sqrt{2}}{K}\left(\frac{3a_\mu}{8} + \frac{3b_\mu}{2N}\right) + 2C_\eta\sqrt{\left(\frac{a_\mu}{6N} + \frac{b_\mu}{2N^2}\right)M_{2,N}},\end{aligned} \tag{26}$$

where $M_{2,N} = \operatorname{Tr}[J_N^2 B_N(I - B_N)] \le 32(H_N+1)/\pi^2$ and $H_N = \sum_{j=1}^N 1/j$. In particular, for fixed $\mu > 0$,

$$\mathcal{Q}_N = \mu^2 c_\eta + O_\mu((\log N/N)^{1/3}). \tag{27}$$

Proof. For the longitudinal part, sum (19) through K, apply the coefficient-sum estimate, replace the symbol using (25), and use the mass-normalization bound. Then add the limiting-series tail and (24). The transverse contribution is the last term of (26): it follows from the covariance pairing inequality and Cauchy–Schwarz,

$$|N\operatorname{Tr}[D_N^{\mathrm{tr}}F(T_N)]| \le \frac{2C_\eta}{N^2}\sqrt{\operatorname{Tr}(G_N J_N^4)\,M_{2,N}}, \qquad \operatorname{Tr}(G_N J_N^4) \le a_\mu N^3/6 + b_\mu N^2/2.$$

These finite inequalities are proved in [8, 9] from the Wilson tail and reference boundary moment. At fixed μ the first term of (26) is $O_\mu(K^2\log N/N)$, the polynomial tail is $O_\mu(K^{-1})$, the symbol remainder is $O_\mu(\log N/N^2)$ and the transverse term is $O_\mu(\sqrt{\log N/N})$. Choose $K = \lfloor (N/\log N)^{1/3}\rfloor$ for sufficiently large N. Every term has the order asserted in (27) or smaller. □

7 Assembly of the entropy finite part

For $N = 2L$ and the exact interval unitary, the scalarized difference $\Delta_N = A_N - B_N$ obeys the finite identity

$$\mathcal{D}_L = \mathcal{P}_L + \mathcal{Q}_N + \mathcal{M}_L - \Delta S_L, \quad \mathcal{P}_L = -\pi \operatorname{Tr}(\Delta_N J_N), \quad \mathcal{M}_L = \operatorname{Tr}(\Delta_N E_N). \tag{28}$$

This follows directly from $\mathcal{D}(A\|B) = \operatorname{Tr}[(A-B)K_B] - S(A) + S(B)$, where $K_B = \log(I-B) - \log B$; it does not require commuting massive and reference covariances. The exact elliptic two-bond calculation (Corollary 4 of [2]) gives

$$\mathcal{P}_L = \frac{\mu^2}{6}\log L + \mu^2\left(\frac{1}{6}\log\frac{8}{\mu} - \frac{1}{4}\right) + O_\mu(\log L/L). \tag{29}$$

To define $\mathcal{S}$, let P^c_μ be the Fourier multiplier on $L^2(\mathbb{R};\mathbb{C}^2)$ with symbol $[I - (u\sigma_x + \mu\sigma_z)/\sqrt{u^2+\mu^2}]/2$ and let $\chi = 1_{(0,1)}$. Put $X_\mu = \chi P^c_\mu(I-\chi)$, $Z_\mu = X_\mu - X_0$, $\mathbb{H}(X) = \left(\begin{smallmatrix} 0 & X \\ X^* & 0 \end{smallmatrix}\right)$, $\mathbb{D}_\mu = \mathbb{H}(Z_\mu)$ and $A_\theta = \mathbb{H}(X_0) + \theta\mathbb{D}_\mu$. These block operators act on the interval and both exterior half-lines. For $C = \chi P^c_\mu \chi$, the projection identity $X_\mu X_\mu^* = C(I-C)$, also at $\mu = 0$, gives $\|A_\theta\| \le 1/2$ by convexity. For $h_2(p) = -p\log p - (1-p)\log(1-p)$ with $0\log 0 = 0$, define

$$H(s) = h_2\left(\frac{1+\sqrt{1-4s^2}}{2}\right), \quad |s| \le 1/2, \qquad \mathcal{S}(\mu) = \frac{1}{2}\int_0^1 \operatorname{Tr}[H'(A_\theta)\mathbb{D}_\mu]\,\mathrm{d}\theta.$$

The finite trace-path lemma and equation (B9) of [10], together with the nuclear estimate of [11], show that $\mathbb{D}_\mu$ is trace class, H' is bounded, and

$$\Delta S_L := S(C_{L,\mu/L}) - S(C_{L,0}) = \mathcal{S}(\mu) + O_\mu((\log L)^{-1}). \tag{30}$$

The trace path is a finite quantity. No subtraction of two divergent continuum entropy traces occurs.

Proof of Theorem 1. Insert (29), (27), (15) and (30) into (28). The constants combine exactly because

$$\frac{1}{6}\log\frac{8}{\mu} - \frac{1}{4} + c_\eta = \frac{13}{36} - \frac{\pi}{12} - \frac{\log\mu}{6}.$$

The errors are, respectively, $O_\mu(\log L/L)$, $O_\mu((\log L/L)^{1/3})$, $O_\mu(L^{-1/2})$ and $O_\mu(1/\log L)$. The last bounds their sum for sufficiently large L, proving (1). □

8 Scope and computational materials

The bound (10) is sublinear. A separate argument in [14] proves $\|E_N\| \le 7$ for every even N, the modular trace rate $O_\mu(\sqrt{\log N}/N)$ and nonlinear rate $O_\mu(\sqrt{\log N/N})$. Those stronger estimates are not inputs to the proof above and leave its full relative-entropy rate unchanged. Neither these bounds nor the entropy theorem supplies a uniform moving-edge Stirling expansion, a general-filling bulk remainder, or a rate uniform in shrinking μ. The rate $1/\log L$ is inherited from the proved boundary approximation and is not an optimized convergence estimate. The entropy comparison is made on the equal-time interval algebra with the stated Wilson regulator. Null-surface comparisons of distinct field theories [13] use a different state identification and cannot be substituted for (28).

The accompanying computational material checks finite-path inequalities with exact rational arithmetic, beta-integral coefficients, scalarization signs, and finite modular and covariance matrices at increased precision. These calculations are independent diagnostics of the analytic formulas; ordinary multiprecision agreement is not an interval certificate.

For a finite-matrix diagnostic, set

$$R_{Q,N} = \mathcal{Q}_N - \mu^2 c_\eta, \qquad R_{D,L} = \mathcal{D}_L - \frac{\mu^2}{6}\log L + \Delta S_L - \mu^2\left(\frac{13}{36} - \frac{\pi}{12} - \frac{\log\mu}{6}\right).$$

Thus $R_{D,L} = R_{P,L} + R_{Q,N} + \mathcal{M}_L$ exactly, where $R_{P,L}$ subtracts the two explicit terms in (29). This diagnostic uses the finite entropy difference and does not numerically approximate the continuum functional $\mathcal{S}$.

μ	L	$R_{Q,N}$	$\mathcal{M}_L$	$R_{D,L}$
0.5	16	3.97015e-05	-2.61926e-05	-0.074798
2	16	-0.000894674	-0.000393623	-0.352826
5	16	-0.0183883	-0.00162353	-1.60106
0.5	32	6.29487e-06	-1.02145e-05	-0.0434371
2	32	-0.000539551	-0.000134702	-0.230774
5	32	-0.0101841	-0.000608888	-1.32463
0.5	64	-6.21958e-07	-3.28072e-06	-0.0247054
2	64	-0.00029966	-4.21723e-05	-0.142284
5	64	-0.00530411	-0.000203647	-0.938197

The full twelve-case data also include $L = 8$ at these three masses. Each case was recomputed at two working precisions; the recorded quantities agree to the 65 significant digits retained in the data files. Values are rounded to six significant digits; trailing zeros are omitted. The small-L normalization was also checked by direct integration of the original two-orbital Bloch symbol and dense matrix logarithms. These checks support the signs and normalizations; they neither determine the asymptotic rate nor certify roundoff by interval arithmetic. At these sizes $R_{D,L}$ is dominated by the known two-bond remainder, which is not required to vary monotonically with L.

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